

## **Data Mining of Amazon Reviews Based on Text Sentiment Analysis, Feature Engineering, and Poison Process**

### **Summary**

The growing demand for people's online shopping leads to growing varieties of products' categories. However, it is difficult for consumers to comprehensively know the quality of a product through its description only. Meanwhile, it is also hard for manufacturers to decide the features their products should obtain. As Amazon creates a platform for consumers to make comments on products, we can know more about how other consumers think about the product. Therefore, we develop several models to figure out the inner relationship between star ratings, reviews and helpfulness ratings and establish a sales strategy for Sunshine Company.

In the first part, we pre-process the data with three rules to delete the incomplete data and tags with no meanings and non-dimensionalize the remaining data. After data cleansing, we apply the term frequency-inverse document frequency (TF-IDF) algorithm to exclude stop words on efficient contents, and figure out the representative words of each product – hairdryer, microwave, and pacifier – and common informative words. Furthermore, we construct a fuzzy system model to refining the rating data and its relation to word semantic as labeled training data to quantization the consumers' review. Hence, we can obtain the relationship between star rating and content of reviews. Then we set up a combination of two key indicators, type of informative word and information entropy of reviews, to assist the Sunshine Company track the market trend.

In the second part, to find out the pattern of how the helpfulness indicators – helpful votes and total votes – from dataset determine the reliability of reviews, we filter out meaningful features that may relate to the helpfulness attribution. For instance, the information entropy of review is a feature we measure. We consider five correlated factors and apply Support Vector Regression to determine the correlation between different impact factors and helpfulness ratings.

In the third part, we notice that historic reviews influence the ones in the future. The poison process matches this pattern to construct a probability density function under given historic events, which has decreasing influence as time flows. Based on this time series analysis, we acquire a variation trend of products' reputation.

Finally, we write a recommendation letter to the Marketing Director of Sunshine Company to expound the analysis of key indicators and features of products, and what sales strategy can we take according to the result of data mining.

**Keywords:** Data Mining, TF-IDF Algorithm, Information Entropy, Support Vector Regression, Feature Engineering, Poison Process, Online Consumer Review

## Contents

|          |                                                                       |           |
|----------|-----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                   |           |
| 1.1      | Background .....                                                      | 2         |
| 1.2      | Problem Restatement.....                                              | 2         |
| 1.3      | Overview of Our Work.....                                             | 3         |
| <b>2</b> | <b>Definitions</b>                                                    |           |
| 2.1      | Assumptions .....                                                     | 5         |
| 2.2      | Definition and Variables .....                                        | 5         |
| <b>3</b> | <b>Data Cleansing</b>                                                 |           |
| <b>4</b> | <b>Text Semantic Analysis Model</b>                                   |           |
| 4.1      | Data groups and relationship .....                                    | 6         |
| 4.2      | Extract keywords with TF-IDF algorithm.....                           | 7         |
| 4.3      | Train emotion dictionary based on fuzzy system .....                  | 9         |
| <b>5</b> | <b>Feature Engineering Model</b>                                      |           |
| 5.1      | Feature filtering .....                                               | 12        |
| 5.2      | Effective information read by users: The calculation of entropy ..... | 13        |
| 5.3      | Determination of the weights of impact factor based on SVR.....       | 14        |
| 5.4      | Design trackable target.....                                          | 14        |
| <b>6</b> | <b>Time Series Analysis Model Based on Poisson Process</b>            |           |
| 6.1      | Influence of historic reviews on future ones .....                    | 16        |
| 6.2      | Mutual information method.....                                        | 18        |
| <b>7</b> | <b>Conclusion</b>                                                     |           |
| <b>8</b> | <b>Strength and Weakness</b>                                          |           |
| 8.1      | Strength .....                                                        | 20        |
| 8.2      | Weakness.....                                                         | 20        |
| <b>9</b> | <b>Recommendation Letter</b>                                          |           |
|          | <b>References.....</b>                                                | <b>23</b> |
|          | <b>Appendix A: Weight of Keywords.....</b>                            | <b>24</b> |
|          | <b>Appendix B: Text Emotion Inclination.....</b>                      | <b>25</b> |

# 1 Introduction

## 1.1 Background

The growing demand for people's online shopping leads to growing varieties of products' categories. The only one defect of online shopping is that it is hard for us to comprehensively know the quality of a product through its description only. Hence, how to distinguish a product's quality becomes a problem. Amazon, as one of the biggest online shopping platform, provides consumers with a review and star rating system, which is to help them take other consumers' comments into consideration for making a better decision.

Meanwhile, to prevent malicious reviews (both good and bad ones), Amazon also allows consumers to vote for other consumers' reviews, judging whether it is helpful or not. Since the review is one of the most critical factors of whether to buy a product, it is quite important to know the inner relationship behind those data of reviews. It can not only help retailers to realize what the possible charming points a product has, but also tell retailers about its popularity and future tendency.

This essay is based on the data of reviews on microwave, hairdryer, and pacifier from Amazon, helping Sunshine Company to research the relationship behind the data, choose the products' best features to be their selling point, and track the selling trend of the market.

## 1.2 Problem Restatement

To find a stable plan for selling products, we need to solve the following problems:

1. Find the inner relationship between star ratings, reviews and helpfulness ratings, and how they synthetically show the extent of consumers' preference on the product.
  - a) Does the emotion inclination of reviews and star ratings match with each other?
  - b) Are there any special words having a higher correlation with star rating?
  - c) Which features among the data have a relationship with helpfulness ratings?
2. Design the most representative index of selling policy based on reviews and star ratings.
3. Research the pattern of how the reputation of a product changes with time.

- a) Find the influence on a product's reputation from a specific combination of reviews and star ratings.
- b) Can a specific ratings in a time duration influence the contents of reviews?

## 1.3 Overview of Our Work

### Part I: Data cleansing and Text Sentiment Analysis

1. **Data Cleansing:** After observation, we find that there occurs some data missing, and meanwhile, some reviews contain non-text tags (e.g. HTML tags). Hence, we need to pre-process the data with python code. We will delete the incomplete data and tags with no meanings, and non-dimensionalize the remaining data.
2. **Text Sentiment Analysis:** To know the relationship between reviews and ratings, since ratings are values while reviews are texts, we apply term frequency-inverse document frequency(TF-IDF) algorithm to analyze the word frequency in reviews, and also exclude the influence of stop words on efficient contents. We divide the given data into the training part and verifying part, and take rating as the tag to create an emotion dictionary for words in reviews. Some circumstances are also considered that the emotion inclination in reviews and ratings are opposite, which can better show the attitudes of reviewers. Hence, basing on the emotion dictionary, we can find whether informative words have a higher correlation with specific star ratings.

### Part II: Feature Engineering

1. **Feature Engineering:** To research the correlative attributes of helpfulness rating, we need to filter out the meaningful features. In the given data, the numbers of rating, review and informative words can be directly approached. Considering that in information theory, the amount of information in a text has a direct relationship with its uncertainty, and also being able to cope with ineffective information created by machine, we take information entropy

and the length of text as a review's feature and apply Support Vector Machine to determine the correlation between different impact factors and helpfulness ratings.

2. **Index designing:** Index can be a combination of ratings and reviews' emotion, or being related to informative words. Based on the model above, we find informative words that can define the quality of different categories of products.

### **Part III: Timing Analysis Based on Poison Process**

1. **Poison process:** According to statistical research on the data, we find that historic reviews influence the ones in the future. Based on the theory of the poison process, we construct a probability density function under given historic events, which has decreasing influence as time flows. Values mapped from the events above are calculated by weighted ratings and reviews. We take the timing analysis based on the poison process and research the pattern of the change of a product's reputation and its specific reason.
2. **Mutual information:** Finally we take mutual information method to research the correlation between rating and informative words in a time duration that reputation of a product changes.

## 2 Definitions

### 2.1 Assumptions

1. Assume that when the emotion inclination in reviews and ratings are opposite, it can better show the attitudes of reviewers;
2. Assume that the majority of reviews are generated by non-machine, which satisfies natural person's logic;
3. Assume that historic reviews have their influence on future reviews, which decreases as time flows.

### 2.2 Definition and Variables

| Notation    | Definition                                     |
|-------------|------------------------------------------------|
| $x$         | Certain attribute of data                      |
| $n$         | Total number of data                           |
| $w$         | Some word in the reviews                       |
| $tf_w$      | Frequency of word $w$ in a group of data       |
| $idf_w$     | Inverse document frequency of word $w$         |
| $n_w$       | Times the word $w$ appears in a group of data  |
| $D$         | Number of efficient reviews                    |
| $r$         | Reliance on helpfulness rating                 |
| $A$         | Fuzzy set                                      |
| $\mu_A(x)$  | Function of fuzzy set $A$ in $x$ 's domain     |
| COG         | Centroid of sample                             |
| $H(review)$ | Information entropy of reviews                 |
| $l(review)$ | Length of texts of reviews                     |
| $p_i$       | The probability that the word $i$ is read      |
| $\lambda$   | The expectation of time that some event occurs |
| $\tau$      | Time length                                    |

### 3 Data Cleansing

The data in the real world is not necessarily perfect and flawless, so online reviews also contain some machine-generated or meaningless reviews. When observing three data sets, we found three main problems and established corresponding processing methods.

1. There are some missing and shifting data in the data sets. For this type of data, we take the treatment of deletion.
2. Some reviews contain HTML tags (i.e. `<br/>`), and we eliminate them by using Python's regular expression method.
3. Because some reviews are not related to the product content, we set up a simple synonym list of some keywords in the product description. When we scan the reviews with the list, we find myriads of irrelevant reviews in pacifier.tsv. Take the review in line 50 as an example, “good cleaning cloths” cannot correctly describe “pacifier”. Processing this type of data requires semantic analysis, so we decide not to process them in the data cleaning part.

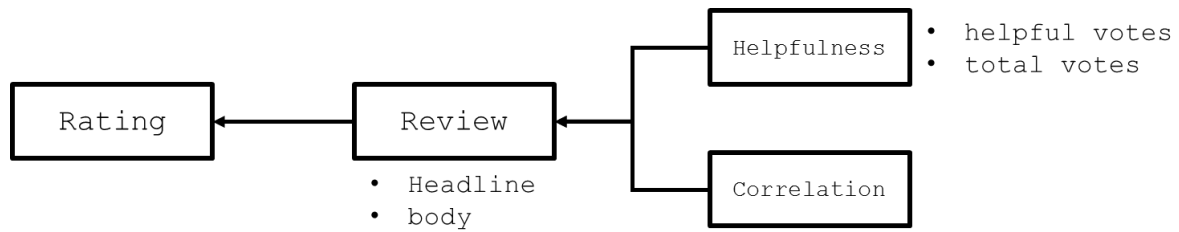
In addition, due to different ranges of the main numerical data of rating, helpful votes, and total votes, we need to unify their specifications. For the subsequent analysis of the model, we choose the zero mean standardized approach. We standardize the data based on Formula 1.  $x$  represents a certain attribute and  $n$  represents the total number of data.

$$x^* = \frac{x - \bar{x}}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}} \quad (1)$$

## 4 Text Semantic Analysis Model

### 4.1 Data groups and relationship

In order to clarify the inner connections among the data, firstly we need to consider how review contents are related to the product quality, which can be measured by rating, review, and helpfulness.



**Fig 1.** The relationship among rating, review, and helpfulness

The value of a comment depends not only on whether the content is consistent with the appraised product but also on whether it is recognized by the public. The former requires semantics analysis while the latter is determined by multiple factors such as the total number of votes and the number of endorsements. The customer review is strongly correlated with rating from the overall perspective of user evaluation behavior, but there are also some cases in which comments and ratings are completely opposite or irrelevant.

## 4.2 Extract keywords with TF-IDF algorithm

First, in order to find qualitative or quantitative relationships between review and rating, we attempt to convert text variables to computable numeric values. TF-IDF algorithm is used as a means to complete a word frequency analysis, which can help to find the keywords strongly related to a product. To consider that the judging standard of the TF-IDF algorithm on whether a word has its representativeness is based on the occasion that the word has a higher frequency of appearance in one group of data while has a lower one in other groups. Hence, our word frequency analysis covers three groups of data. Suppose some word  $w$  has its frequency  $tf_w$ , and its inverse document frequency is  $idf_w$ , then there exists the relationship as the following :

$$tf_w = \frac{n_w}{\sum n} \quad (2)$$

Where  $n_w$  indicates how many times the word appears in the group of data, and  $\sum n$  indicates the whole number of words in the group of data.

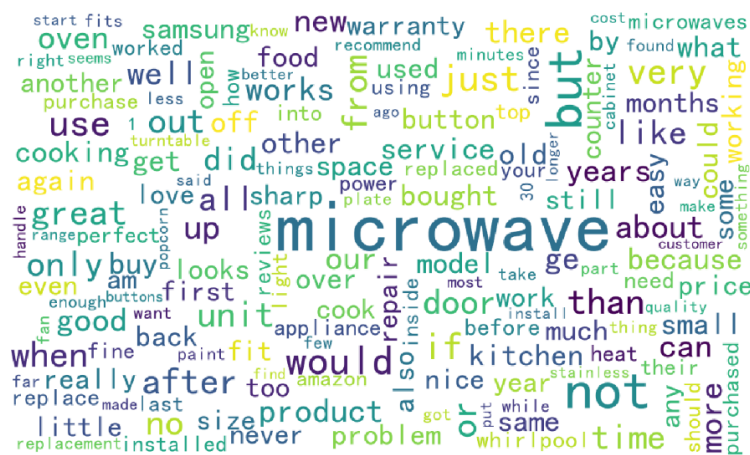
$$idf_w = \log \frac{|D|}{|i: w \in d_i| + 1} \quad (3)$$



Where  $D$  is the number of efficient reviews,  $|i: w \in d_i|$  represents the number of reviews that contain the word  $w$ . Finally, by multiplying these two factors, we can assess the importance of the word:

$$\text{TF-IDF} = \text{tf}_w \times \text{idf}_w \quad (4)$$

Processing it on Python and depicting word cloud as follows.

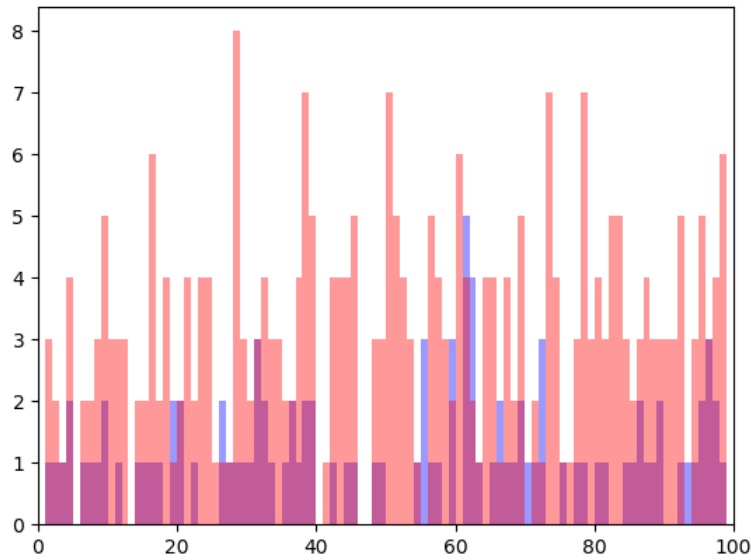


**Fig 2.** Word cloud of microwave.tsv

Among those reviews, there exist some stop words which have high frequency but no meaning, which have an influence on effective content. Although the TF-IDF algorithm can filter out the majority of stop words, because of the limited scale of samples, it is unable to filter out all of them. For words like “did”, we do a manual reprocess, and finally get the following table which shows representative words of each product:

**Table 1.** Representative words of each product

| Product      | Informative Words                  |                  |
|--------------|------------------------------------|------------------|
|              | Positive                           | Negative         |
| Hairdryer    | time, light, year, powerful, price | blow, cord, stop |
| Microwave    | unit, work, kitchen                | door, only, out  |
| Pacifier     | baby, little, easy, cute           | mouth, no        |
| Common Words | great, nice, fine                  | not, but         |



**Fig 3.** Histogram of common and particular informative works (Ranked by weight)

The orange color indicate the particular words, and the blue part refers to the common words. We can see that the frequency of common informative words is lower than the particular one. The result of the calculation on the weight of keywords is stated in Appendix A.

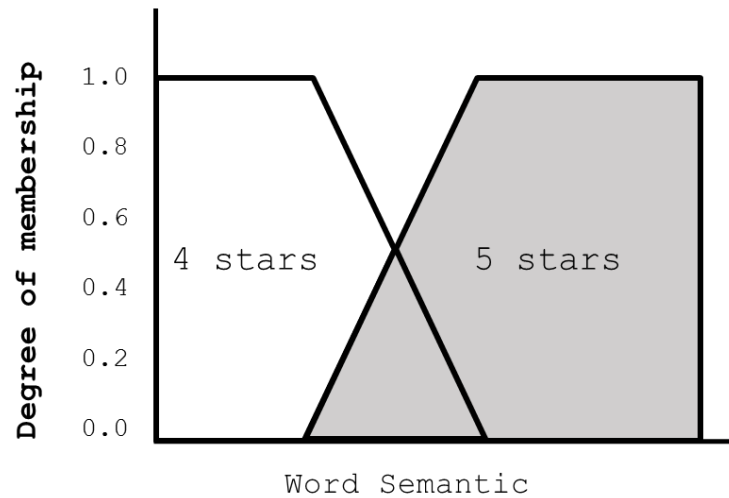
### 4.3 Train emotion dictionary based on fuzzy system

We first assume that the reliance of helpfulness rating  $r = \frac{\text{votes}_{\text{helpful}}}{\text{votes}_{\text{total}}}$  is relatively high if the review is reliable. We divide the data into groups for training and group for verifying. The Training group contains reviews with higher reliance, while the verifying group contains reviews with total votes equal to 0. Those reviews maybe not unreliable, but only not be voted. Then we take the value of rating as the tag to simply train the emotion dictionary of words in reviews.

In terms of how to connect the rating with word's emotion, since there are only 5 ranks in rating system, while the emotion inclination some specific words convey may be more than 5 ranks, we hence construct fuzzy system which can preset degree of membership of words and get the result through inverse gelatinization, and finally generate the emotion dictionary for the given data.

Let data's domain  $x$ 's fuzzy set  $A$  has its function  $\mu_A(x)$ , then  $\mu_A(x):X \rightarrow [0,1]$ 。 Applying AND fuzzy process to the input and we get :

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x)) \quad (5)$$



**Fig 4.** Clear and fuzzy sets of word-star rating

We apply the centroid technology to accomplish the inverse gelatinization. The calculation for the sample's centroid COG is:

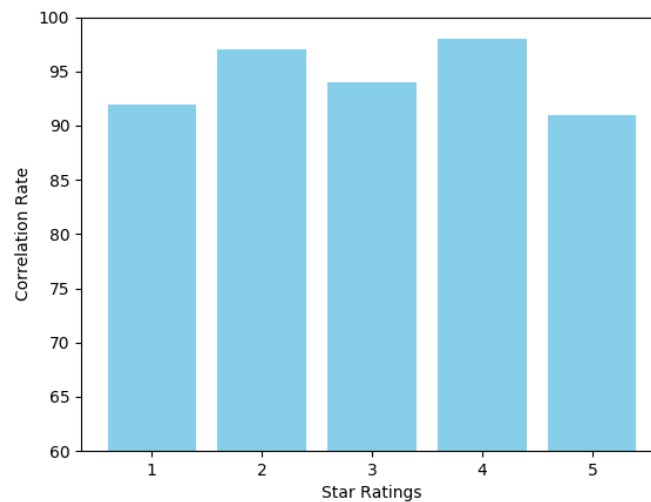
$$\text{COG} = \frac{\sum_{x=a}^b \mu_A(x)x}{\sum_{x=a}^b \mu_A(x)} \quad (6)$$

The result of different texts' emotion inclination (we choose 100 lines) is listed in Appendix B. We map it into the range of  $[0,1]$  and the following table shows part of the reviews and their related values of inclination based on the fuzzy system.

**Table 2.** Quantize the emotion inclination of reviews (hairdryer)

| ID  | Review                                                    | Keyword  | Value       |
|-----|-----------------------------------------------------------|----------|-------------|
| 90  | great for the price!! ..... I love it!!                   | price    | 0.854166667 |
| 112 | this is one of the best hair dryer .....win all around    | powerful | 0.8125      |
| 12  | the styling tip does not stay on, ... too hot to put back | blow     | 0.083333333 |

Based on the data above, we study whether those reviews match the ratings, and the result is shown as follows.



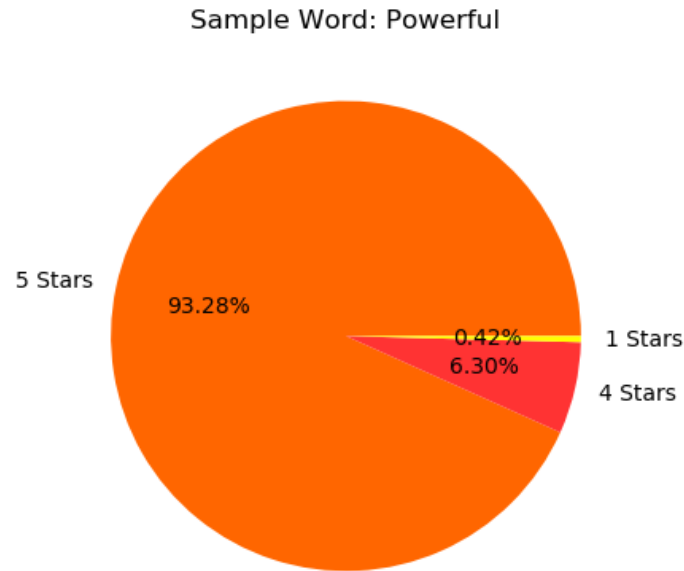
**Fig 5.** Consistency between reviews and ratings among the data

Some circumstances do exist that the emotion inclination in reviews and ratings are opposite, which can better show the attitudes of reviewers. Hence, we cannot simply bind those words with star ratings. Instead, basing on the emotion dictionary, we find out some informative words do have a strong relationship between specific star ranks. The correlation between star ranks and specific keywords is shown below:

**Table 3.** Correlation of star ranks and specific keywords

| Rating | Informative Words                     |
|--------|---------------------------------------|
| 5      | perfect, great, nice, all, easy, work |
| 4      | fine, well, only, even, enough        |
| 3      | other, just                           |
| 2      | (No typical informative words)        |
| 1      | not, but, no                          |

We also find that the average scores and variance of these informative words differ a lot with the whole dictionary. Informative words have smaller variance, which meets the real circumstance that they have higher representativeness.



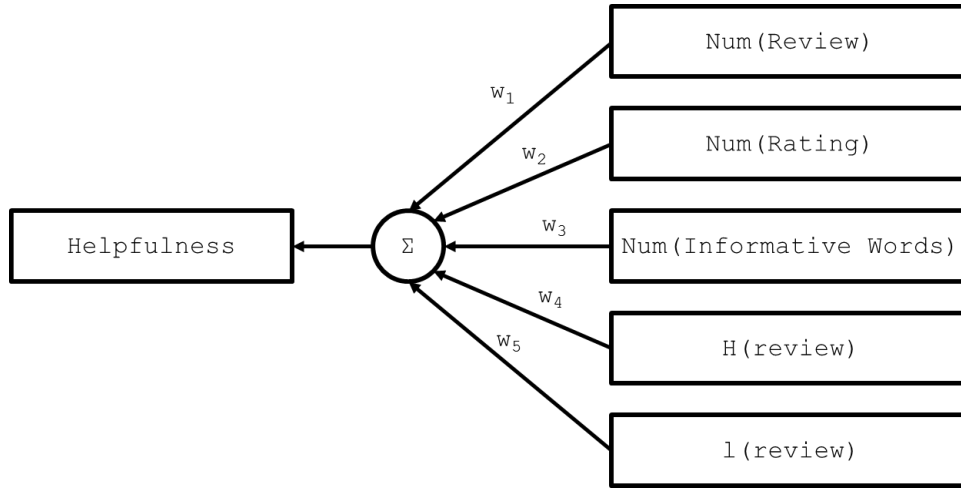
**Fig 6.** Sample of word that has strong/little relationship with some specific star rank

## 5 Feature Engineering Model

### 5.1 Feature filtering

In the text semantic analysis model, we first construct a semantic inclination model, find representative informative words and study the correlation between reviews and ratings. This model is designed to have a further study of how the value of helpfulness rating can define a review's quality, which means that we need to filter out the features related to helpfulness rating.

When we choose a meaningful feature, it needs to be divergent, with variance not close to 0. All the types of data meet the demand. In the given data, the numbers of rating, review and informative words can be directly approached. Considering that in information theory, the amount of information in a text has a direct relationship with its uncertainty, and also being able to cope with ineffective information created by machine, we take information entropy  $H(review)$  and the length of text  $l(review)$  as a review's feature, and apply Support Vector Machine to determine the correlation between different impact factors and helpfulness ratings.



**Fig 7.** Association graph that has relation to helpfulness

## 5.2 Effective information read by users: The calculation of entropy

The amount of effective information determines the value of a message. For the product reviews on Amazon, the information truly read by the other users is the effective information from the words excluding the "stop words", which have been discovered in the previous models. We calculate the entropy for every comment with the method mentioned in Shannon's information theory:

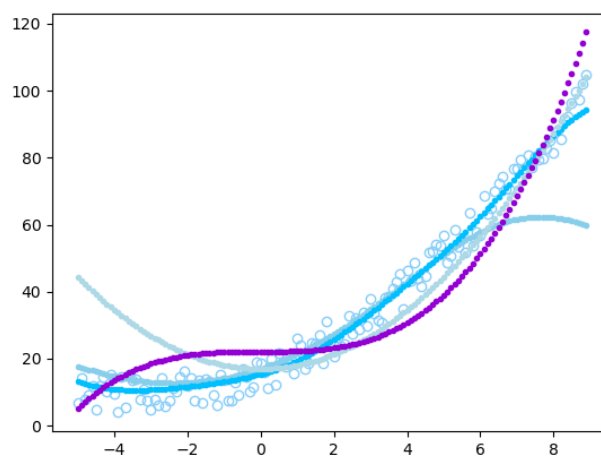
$$H(\text{review}) = - \sum_{i=1}^n p_i \log p_i \quad (7)$$

in which  $i$  represents the words in a comment,  $p_i$  represents the probability that this word is read. Considering 99% of reviews in the data set have less than 50 non-stop-words, we believe this probably won't decrease with the comments being hidden further as the long articles do. Here, we take the frequency the word occurs as this probability.

In addition, it is easy to obtain the range of  $H(\text{review})$  is  $0 \leq H(\text{review}) \leq \log n$ .

### 5.3 Determination of the weights of impact factor based on Support Vector Regression

Since the data specification of five impact factors are still different, we standardize and normalize the data again within the method of formula (1). Furthermore, we utilize the SVR as a classifier, which is often seen in feature engineering. We apply the SVM method in python's sklearn machine learning library to determine the weights. Noted that the fitting results may only be effective for given data sets due to fewer sample data as well.



**Fig 8.** Support vector regression determines the weight of the impact factor

As a result of our analysis, we indicate the entropy of Helpfulness and reviews are closely related to whether they have informative words mentioned above. From this conclusion, we can design tracking indicators in market research for Sunshine Company.

### 5.4 Design trackable target

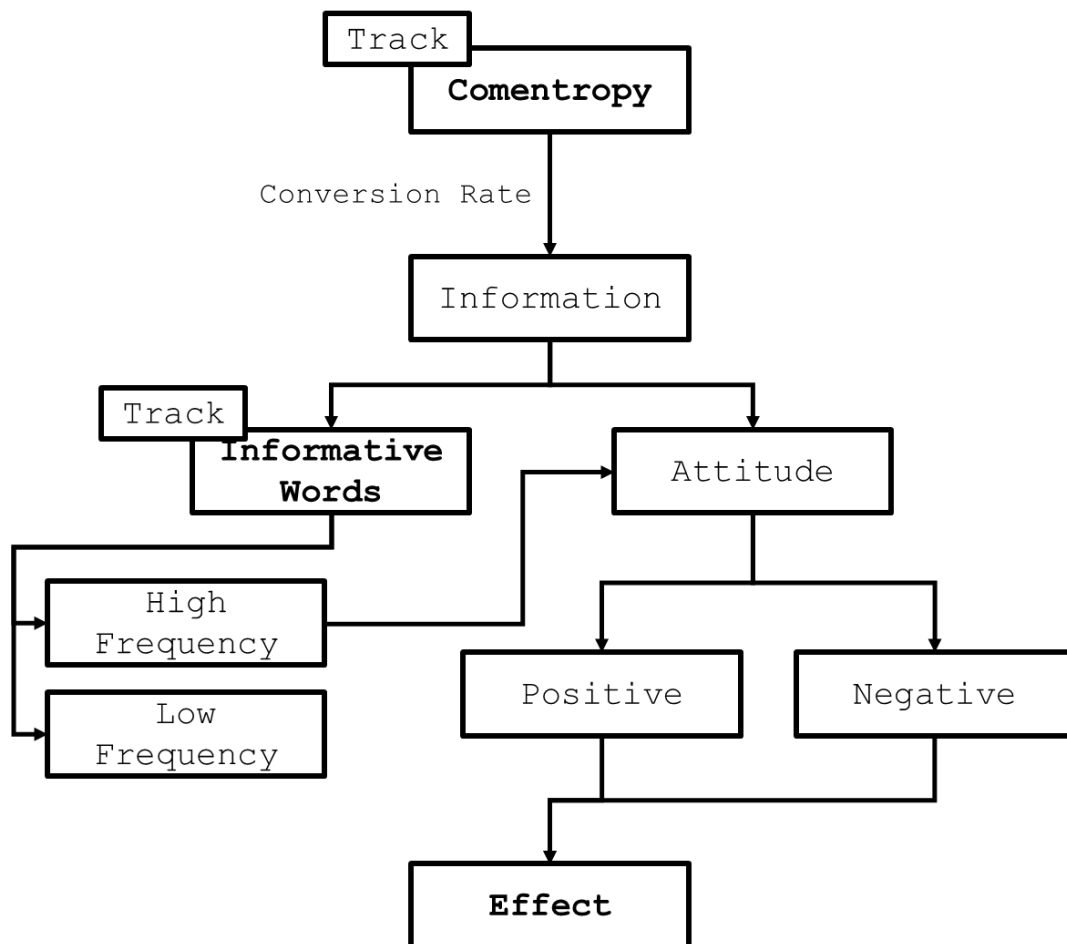
According to the models above, we choose the following targets :

1. Type and number of informative words over a period of time
2. Average information entropy of reviews over a period of time

According to the statistic analysis and logic, information entropy determines whether consumers can better get effective information from the contents of reviews so that they can have higher attention to the product. This indicates the transformation ratio from text information to getting information, while the number of informative words determines the most

influencing information on consumers' decisions among the remaining information they get. The degree of positive/negative mood of these words can have a great influence on a product's reputation.

Then we design the market tracking flow chart based on the targets and analysis above for Sunshine Company:



**Fig 9.** Market Tracking Flow Chart

**Table 4.** Circumstances explained by the combination of targets

| Informative Words<br>Attitude | High Frequency | Low Frequency |
|-------------------------------|----------------|---------------|
| Positive                      | Good Condition | Normal        |
| Negative                      | Attention      | Normal        |



## 6 Time Series Analysis Model Based on Poisson Process

### 6.1 Influence of historic reviews on future ones

According to the statistic analysis on data, we find that the reviews in the past have an influence on future reviews. In general cases, the website of Amazon shows the recent reviews in priority, which means that the influence of historic reviews on future ones decays as time flows. The trend that the reputation of a product changes with time matches the Poisson process. In the Poisson process, history is the probability density function of future events, and its influence decreases as time goes.

We define the event as one rating behavior from a consumer. Suppose over a time period that appears a specific number of reviews, the probability of existing specific times of different ratings is  $P(x = k)$ , then we build the following model :

$$P(x = k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad (8)$$

Where  $\lambda$  is the expectation of time that some event occurs.

For example, after observation and analysis, we find that a product can get 10 5-star reviews per day. Then, to calculate the probability that the product will get 3 5-star reviews in the next day, we can apply the following formula:

$$P(x = 3) = \frac{10^3 e^{-10}}{3!} \quad (9)$$

Furthermore, if we need the probability of getting 3 to 7 5-star reviews, the formula can be expanded as:

$$P(3 < x < 7) = \sum_{x=3}^7 \frac{\lambda^x e^{-\lambda}}{x!} \quad (10)$$

Then we get the general form:

$$P(a < x < b) = \sum_{x=a}^b \frac{\lambda^k e^{-\lambda}}{k!} \quad (11)$$

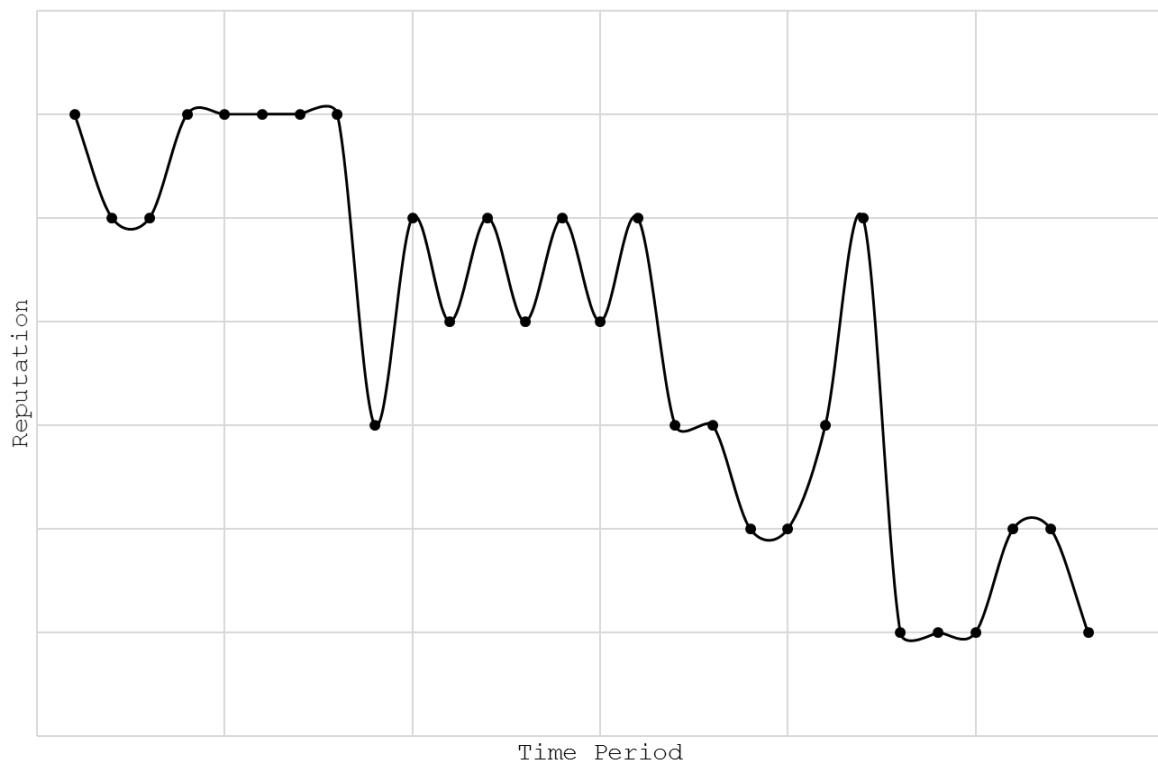
After applying the time period, let  $\tau$  be the time length, then the formula expanded based on time series can be:

$$P(N(t + \tau) - N(t) = k) = \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!} \quad (12)$$

And we can get the equation for the relationship between time and the ratio of good reviews :

$$P(a < N(t + \tau) - N(t) < b) = \sum_{(N(t+\tau)-N(t))=a}^b \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!} \quad (13)$$

After calculating the data and depict the graph, we finally get the changing pattern of hair dryer's reputation as follow:



**Fig 10.** Change the pattern of hair dryer's reputation (B00TA1JX3A as sample)

Among them, we can find there are some turning points that reputation changes. These time periods can help us study whether rating can influence the reviews.

## 6.2 Mutual information method studying the relationship between ratings and subsequent reviews

Basing on the time series constructed by the Poisson process, we apply mutual information method to study the relationship among ratings, the number of subsequent reviews, and the inclination of informative words in the specific time period when the reputation of a product changes. Based on the concept of information entropy in formula (7), we construct pointwise mutual information to judge the correlation between numbers of reviews and the words used in this time period.

$$PMI(x; y) = \log \frac{p(x, y)}{p(x)p(y)} \quad (14)$$

At this moment, if  $x$  is uncorrelated with  $y$ , then  $p(x, y) = p(x)p(y)$ , reversely,  $p(x, y) \gg p(x)p(y)$ . Also to simplify the calculation and increase the efficiency of programs, formula (14) can be simplified as:

$$PMI(x; y) = \log \frac{p(y|x)}{p(y)} \quad (15)$$

According to the data, we get the qualitative result: ratings have some relationship with subsequent reviews. Take the product B00TA1JX3A of microwave.tsv as an example, the reasons for the sudden decrease during the time period 5/25/2015 to 8/8/2015 can be:

1. Reviews:
  - a) Text: “Nice product, but never got the diffuser it was supposed to come with.”
  - b) Rating: 3 stars
  - c) Votes: 1 helpful / 1 total
2. Keywords in new reviews: “blow”, “stop”
3. Numbers of new reviews: 3 (1 of one star, 2 of two stars).

## 7 Conclusion

In this paper, we set up several models to analyze the relationship between star ratings, reviews and helpfulness ratings, and figure out what features of products do consumers interested in, and how can we learn the trend of a product's reputation based on reviews and star ratings from consumers.

To question 1, we use the TF-IDF algorithm to extract informative words from datasets and find out the representative words of each product. From these words, we find out some features of the products, for instance, a powerful hairdryer is welcomed among consumers. Furthermore, we established an emotion dictionary based on the fuzzy system, which is used to analyze the semantic of words and sentences. We figure out that most of the emotion inclination of reviews and star ratings match with each other, however, the correlation of 1 star, 3 stars, and 5 stars are lower than the other two.

To question 2, we use the feature engineering method to determine the impact factors of a review's reliability and find out the type and number of informative words and information entropy of reviews can be the combination of key indicators for the Sunshine Company to track the market trend. We also find out that words like "perfect" do have a close relation to specific star ratings. As an example, the word "powerful" is relevant to a 5-stars rating for a probability of 93.28%.

Last but not least, to question 3, we set up a time series analysis model based on the Poisson Process to understand the pattern of how the reputation of a product changes with time. We discovered that the combination of a representative word we listed before and the relevant level of star rating can influence the contents of reviews.

Finally, we draw 3 figures of word cloud to represent the weight of keywords of three products which can assist the Sunshine Company to make better decisions.

## 8 Strength and Weakness

### 8.1 Strength

1. **Data training:** In order to understand the relationship between reviews and ratings, we apply text sentiment analysis based on the term frequency-inverse document frequency (TF-IDF) algorithm. Differ from other sentiment word dictionary that based on a text train set with wider range, we built up a sentiment word dictionary that only aim on the review data from Amazon, which means it can represent sentiment trend of costumer's comment better; meanwhile, it can assist to erase stop words in a high accuracy.
2. **Informative words:** Fuzzy system is applied in order to handle the non-numerical value of the review. Furthermore, we find out informative words with more quantities of information according to the variance of quantized value. In this way, we provide an easy and accurate method for companies to track a related index of reviews.
3. **Time-series:** We build up the time series analysis based on probability density instead of real-time. This can help us to be more sensitive to the analysis of rating events. Under this condition, we can find out the point of inflection as well as change the mode of reputation mode accurately, which can help us to make fairly reliable forecasts as well as strategies.

### 8.2 Weakness

1. Label used in the process of data training is rather rough, which may bring negative effects to the result of training.
2. Limited amount of data leads to relatively low reliance on machine learning. We applied machine learning algorithm during the construction of models, which are used to train the emotion dictionary of words. The amount of data, especially the one of pacifier, is far from enough, which makes it unclear whether it has the required statistical sense to be popularized. However, it is appropriate enough under the given data.

## 9 Recommendation Letter

**Mailto:** Sunshine Company

**Date:** March 9, 2020

**Subject:** Sales strategy based on data mining of Amazon reviews

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Dear Marketing Director of Sunshine Company,

Based on your demand for seeking the relationship of star ratings, reviews and helpfulness ratings, the important design features of the three products – microwave, pacifier, and hairdryer, and the sales strategy, we established several models to give you some recommendations.

Firstly, with the help of text semantic analysis, we figure out that there exist some informative words which are representative words of each product:

| Product      | Informative Words                  |                  |
|--------------|------------------------------------|------------------|
|              | Positive                           | Negative         |
| Hairdryer    | time, light, year, powerful, price | blow, cord, stop |
| Microwave    | unit, work, kitchen                | door, only, out  |
| Pacifier     | baby, little, easy, cute           | mouth, no        |
| Common Words | great, nice, fine                  | not, but         |

For instance, if a hairdryer is **powerful**, it is much easier for consumers to use, and it will actually take less **time**. This case is logical and easy to explain. Additionally, about the microwave, it is hard to think about how can the word **door** reflect the rating, however, most of the 1-star rating indicated that there is something wrong with the design of the opening or closing the door of the microwave oven. Therefore, while designing the products, make sure to think about these informative words. We can learn what we should do from the positive words, and we can understand how to improve with the help of other negative words.

In addition, we find out the type and number of informative words and information entropy of reviews can be the combination of key indicators for you to track the market trend. In detail, if the appearance of informative Words is frequently, we need to notice the attitude of these

words. If they are positive, it may refer that your product is selling in good condition. Otherwise, you need to take care about whether there are negative comments or not.

Also, we find out that words like “perfect” do have a close relation to specific star ratings. As an example, the word “powerful” is relevant to a 5-stars rating for a probability of 93.28%. In this case, it is a good indicator to show whether a product has a good reputation or not.

Moreover, our recommendation on the sales strategy is, while finding the combination of negative informative words we provided to you in the appendix, such as the keywords “blow”, “stop” of the hairdryer product, there will exist a risk of the decrease of product’s reputation. However, if we are able to publish some reviews by ourselves or vine users timely, the trend of decreasing will be suspended or discontinued. We believe that the consumer’s review impacts a lot on the sales volume as a result. Therefore, please pay close attention the the key indicators we mentioned above.

Finally, our analysis is entirely based on the processing of the dataset your company provided, but not our subjective speculate or irrelevant information. We hope that you can find these suggestions useful to you, and we sincerely wish your success in the products selling on Amazon! Please feel free to contact us anytime if you have any questions or concerns.

Sincerely yours,

Team #2002531

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## Appendix B: Text Emotion Inclination (Source: Hairdryer)

| ID  | Review             | Estimated Value |
|-----|--------------------|-----------------|
| 100 | over heated in ... | 0.35            |
| 101 | nice quality....   | 0.6             |
| 102 | it works great ... | 0.8             |
| 103 | i only had the ... | 0.083333        |
| 104 | i purchased the... | 0.3             |
| 105 | works great ! l... | 0.8125          |
| 106 | this thing is s... | 0               |
| 107 | the speed and h... | 0.015           |
| 108 | so far so good...  | 0.4             |
| 109 | nice quality....   | 0.6             |
| 110 | works just fine... | 0.416667        |
| 111 | excellent hair...  | 1               |
| 112 | this is one of ... | 0.451111        |
| 113 | i love this lit... | 0.087879        |
| 114 | holy cow...mayb... | 0.048437        |
| 115 | the dryer itsel... | 0.55            |
| 116 | cool speed is p... | 0.675           |
| 117 | no warm setting... | 0.46875         |
| 118 | i wish it was a... | 0.07            |
| 119 | it's good. i li... | 0.7             |
| 120 | great blow drye... | 0.1625          |
| 121 | i love my hot p... | 0.269583        |
| 122 | good hair dryer... | 0.215           |
| 123 | excellent produ... | 1               |
| 124 | this was a grea... | 0.042857        |
| 125 | i bought this f... | 0.2             |
| 126 | excellent hair ... | 0.342273        |
| 127 | i just love it...  | 0.5             |
| 128 | love how light ... | 0.438889        |
| 129 | this dryer work... | 0.090139        |
| 130 | buttons interfe... | 0               |
| 131 | from ms. s1rude... | 0.196835        |
| 132 | works great!! j... | 1               |
| 133 | traveling abroa... | 0.025           |
| 134 | love love love ... | 0.625           |
| 135 | my boyfriend us... | 0.45            |

|     |                    |          |
|-----|--------------------|----------|
| 136 | i use it on my ... | 1        |
| 137 | cons: i took a ... | 0.0225   |
| 138 | awesome dryer! ... | 0.258333 |
| 139 | i bought this h... | 0.183333 |
| 140 | i just love it!... | 0.625    |
| 141 | god product...     | 0        |
| 142 | does not turn o... | 0        |
| 143 | it stays true t... | 0.35     |
| 144 | this was for 3 ... | 0.35404  |
| 145 | great hair drye... | 0.583333 |
| 146 | love this hair ... | 0.5      |
| 147 | it's a good, no... | 0.086148 |
| 148 | my wife loves t... | 0.2      |
| 149 | love this hair ... | 0.625    |
| 150 | best dryer i ha... | 1        |
| 151 | junk... the dif... | 0.025    |
| 152 | i used this blo... | 0.109375 |
| 153 | very nice hair ... | 0.8275   |
| 154 | did not work in... | 0        |
| 155 | gets the job do... | 0        |
| 156 | good!...           | 0.875    |
| 157 | dryer works gre... | 0.275    |
| 158 | quite compact...   | 0.255833 |
| 159 | it is everythin... | 0.1      |
| 160 | works great...     | 0.8      |
| 161 | shipment takes ... | 0.43     |
| 162 | i have had alot... | 0.08     |
| 163 | bought as a rep... | 0.1      |
| 164 | this is a nice ... | 0.162889 |
| 165 | not use yet, bu... | 0        |
| 166 | woks good...       | 0.7      |
| 167 | works very well... | 0.016667 |
| 168 | conair always m... | 0.7      |
| 169 | great size and ... | 0.566667 |
| 170 | small, compact,... | 0.0625   |
| 171 | i want you to s... | 0.242424 |
| 172 | i really wanted... | 0.201712 |
| 173 | i bought this h... | 0.066667 |
| 174 | the wire mesh o... | 0.033889 |

|     |                    |          |
|-----|--------------------|----------|
| 175 | it's nice. i th... | 0.247917 |
| 176 | the fan speed i... | 0.01     |
| 177 | the hair dryer ... | 0.1      |
| 178 | i just really l... | 0.064583 |
| 179 | junk! highly no... | 0.125    |
| 180 | i'm on the road... | 0.300152 |
| 181 | my hairdryer wa... | 0.150182 |
| 182 | i love it this ... | 0.5      |
| 183 | this is a great... | 0.287407 |
| 184 | i already have ... | 0.263247 |
| 185 | this is a very ... | 0.255007 |
| 186 | worked great fo... | 0.4      |
| 187 | haven't gotten ... | 0.016667 |
| 188 | great quality....  | 0.8      |
| 189 | let me start wi... | 0.267515 |
| 190 | love it! just d... | 0.625    |
| 191 | it's a bit heav... | 0.408333 |
| 192 | excelente!!!!...   | 0        |
| 193 | good....           | 0.7      |
| 194 | needed a new dr... | 0.179305 |
| 195 | loved!!! my hai... | 0.441667 |
| 196 | i never realize... | 0.170833 |
| 197 | not as strong a... | 0.2      |
| 198 | love it. finall... | 0.229388 |
| 199 | i used one of t... | 0.291667 |